Roberto Pastor Muela

Software Developer

25 April 1995

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RobPasMue

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About me

Aerospace engineer with a main focus on SW development. My main interests are related to numerical simulations, CFD, flight dynamics systems and astrodynamics. SW architect and developer in computationally-intensive programming languages. Great interest on algorithms and the physical knowledge behind them.

Language Skills

Spanish:	Mother tongue
English:	C1*
German:	B2*
French:	A2*

**According to the Common European Framework of Reference for Languages (CEFR)*

Programming Skills

C, C++, CUDA

Fortran

Python

Java

MATLAB

Experience

2022/- Senior R&D Engineer — *ANSYS*

Development of ANSYS SW products and supporting systems. Open-source PyAnsys libraries development. Technological transformation of business units. Software architect for Python projects, driven by usability, customer feedback and automation. GitOps lead for PyAnsys ecosystem teams. Code best-practices and tools guidelines (focused on Python and C/C++). Provided guidance on client-service interactions using REST and gRPC protocols. Product componentization using Docker and service deployment on Kubernetes clusters. Conception of a developer portal PoC using Backstage.

2019/22 Flight Dynamics (FD) and Operations engineer — *GMV*

Precise orbit determination (POD) engineer in projects for institutional clients - Copernicus POD Service (for Sentinels 1, 2, 3 & 6). Experience in FD softwares (SWs) - NAPEOS, SIRIUS, Orekit, GODOT. SW architect and developer of in-house FD library in C++. CI/CD Responsible. Working with Agile methodologies. Knowledge on GNSS systems and signals processing. Design & impl. of API REST servers.

2017 Processing, automatization and development — *Airbus Def. & Sp.*

Development & automatization of qualification tests for rotary actuators used in antenna pointing mechanisms. Post-processing of data to determine system capabilities and requirements verification.

2014/16 Teaching Professional and Camp Coordinator — *Leisure Companies*

Coordinator for children ranging 4 to 14 years old. Entirely taught in English. Team managing capabilities and communication skills.

Education

2017/19 M.Sc. Aeronautical Engineering — *UC3M*

Focus on CFD, space systems and numerical simulations

Master thesis: *Analysis and control of turbulent skin friction using GPU-DNS*

2015/16 B.Sc. Aerospace Engineering — *Universität Stuttgart*

ERASMUS+ Exchange program

2013/17 B.Sc. Aerospace Engineering — *UC3M*

Minor in Aerospace Propulsion

Bachelor thesis: *Improvement of real-time plasma diagnostics*

Additional Courses

2019	Java Programming Masterclass for SW Developers	80h
2019	C++ Basics and Introduction to C	25h
2018	Design thinking course at UC3M	23h
2018	AIRBUS: Innovative thinking	12h
2015	CATIA V5-6: An introduction to CAD	40h

Awards

2015/16 ERASMUS+ Exchange Program scholarship - 1 full academic year at the Universität Stuttgart

2013/18 Modules with honours: Propulsion Systems, Engineering Graphics, Advanced Mathematics, Fluid Mechanics, Thermal Engineering (6 ECTS each) and Aerospace Project Management (3 ECTS)

2013/15 *"Excellent academic record"* scholarship for university students

Interests

I am a very active person who performs regularly multiple team sports, such as football and rugby. I also enjoy travelling and getting to know different cultures, since I consider it a mind-opening experience. I also enjoy living abroad (+6 years over my life).

CV was last updated on March 12, 2024